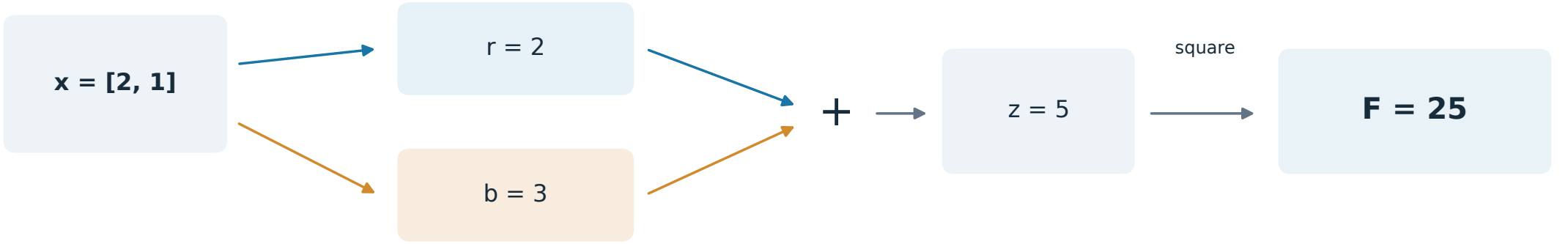
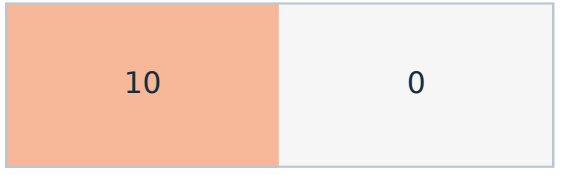
Residual branches: count each route once

Analytic example: $r = x_1$, $b = 2x_1 - x_2$, $z = r + b$, final target $F = z^2$.



Along $x(\alpha) = [2\alpha, \alpha]$: $z = 5\alpha$, $\partial F/\partial z = 10\alpha$.

Via the skip edge

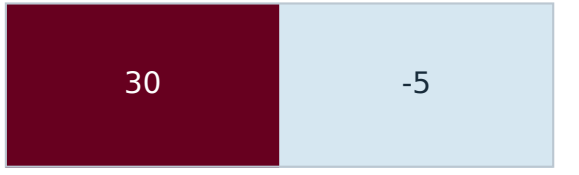


+

Via the computed branch

=

Learning image for F



$10 + 20 - 5 = 25 = F(x) - F(0)$

Branch-alone response map: $[4, -1]$. Its contribution to F is $[20, -5]$, including the downstream derivative.